



Profit-Driven Decision Analytics for Telecommunications Customer Retention: A Cost-Sensitive Stacking Ensemble Framework Optimized by an Improved Sparrow Search Strategy Algorithm

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Abstract: Customer retention in the telecommunications industry poses a critical challenge in data-driven business operations, while high predictive accuracy does not necessarily translate into superior commercial outcomes under asymmetric misclassification costs. This study investigated a profit-oriented decision analytics framework for customer churn management by integrating predictive performance with business value optimization. A cost-sensitive Cost-Sensitive Improved Sparrow Search Strategy Algorithm Stacking (CS-ISSA-Stacking) framework was developed by incorporating customer lifetime value (CLV) and marketing intervention costs into an Expected Total Profit (ETP) objective function. An Improved Sparrow Search Algorithm (ISSA) was constructed using Tent chaotic mapping and Cauchy mutation mechanisms to enhance global optimization capability. EXtreme Gradient Boosting (XGBoost), Light Gradient Boosting Machine (LightGBM), and Categorical Boosting (CatBoost) were employed as heterogeneous base learners, and ISSA was used to optimize ensemble fusion weights and decision thresholds for profit-driven prediction. The proposed framework could effectively improve ETP while maintaining competitive predictive performance. Compared with conventional prediction models optimized solely for classification accuracy, the proposed approach achieved a substantial improvement in customer retention profitability. The findings demonstrated that the optimized decision threshold significantly reduced the false negative rate in high-value customer identification. Furthermore, SHapley Additive exPlanations (SHAP)-based interpretation revealed that monthly charges, contract type, and tenure were the most influential factors affecting customer churn behavior. Profit-sensitive decision mechanisms provided a more effective strategy than traditional accuracy-oriented prediction approaches in customer churn management. The proposed framework provides a practical decision-support tool for intelligent customer retention and offers new insights into integrating business analytics with strategically operational decision making in the competitive telecommunications environments.

Keywords: Telecommunications; Decision analytics; Cost-sensitive learning; Customer retention; Profit optimization; Stacking ensemble

1 Introduction

As the telecom market enters an era of stock competition, preventing customer churn and implementing refined operations have become crucial strategies for operators to maintain core profitability. In recent years, advanced machine learning techniques such as ensemble learning have demonstrated outstanding classification performance in telecom churn prediction [1]. For instance, gradient boosting-based algorithms including eXtreme Gradient Boosting (XGBoost) [2] and Light Gradient Boosting Machine (LightGBM) excel in processing high-dimensional sparse features [3, 4]. However, most existing cutting-edge churn prediction studies take maximizing accuracy or Area Under Curve (AUC) as the single optimization objective [5, 6]. This orientation generally relies on the assumption of "equal misclassification costs". In real commercial risk control scenarios, the long-term revenue loss caused by missing high-value customers (false negatives) far exceeds the marketing retention cost wasted by misclassifying low-value customers (false positives). Such a severe disconnect between technical metrics and business objectives often prevents high-score prediction models from maximizing the Expected Total Profit (ETP) desired by enterprises in practical deployment.

To address the above asymmetric misclassification cost, cost-sensitive learning has been introduced into the field of churn prediction [6–8]. Nevertheless, in existing research, systematic studies that deeply integrate specific customer lifetime value (CLV) into the cost matrix and conduct instance-level profit evaluation remain relatively scarce [9]. Furthermore, when constructing advanced ensemble prediction frameworks with stronger generalization ability (e.g., the stacking mechanism) [10], the optimization of base model fusion weights directly determines the upper limit of the model’s business transformation [11]. As a novel swarm intelligence optimization algorithm, the Sparrow Search Algorithm (SSA) [7, 12] displays certain advantages in local search and global exploration. However, when facing the extremely high-dimensional and highly discontinuous profit hyperplane, the standard SSA is prone to bottlenecks such as declining population diversity and premature convergence [13], thus urgently requiring the introduction of more advanced mathematical mechanisms for improvement [14, 15].

In view of this, this paper proposed a cost-sensitive stacking ensemble prediction framework based on an improved Sparrow Search Optimization Algorithm, namely Cost-Sensitive Improved Sparrow Search Strategy Algorithm Stacking (CS-ISSA-Stacking). This study broke the assumption of equal misclassification costs in traditional models and reconstructed an instance-level profit objective function combining individual CLV and marketing costs [16]. Meanwhile, Tent chaotic mapping and Cauchy mutation strategies were introduced to Improved Sparrow Search Algorithm (ISSA) [6, 17], thus enhancing the algorithm’s global dynamic optimization capability. Finally, a stacking architecture was constructed with XGBoost, LightGBM and Categorical Boosting (CatBoost) [18] as heterogeneous base learners, and ISSA was used to adaptively search for the optimal fusion weights and classification decision thresholds. Aiming at highly imbalanced telecom data scenarios, this paper effectively bridged the gap between prediction accuracy and commercial economic benefits, so as to provide profit-driven intelligent decision support for a wider range of commercial risk control fields [19, 20].

2 Theoretical Foundation and Model Construction

2.1 Heterogeneous Base Learners and Stacking Integration Strategy

When dealing with telecom customer churn data with high-dimensional sparse and complex nonlinear features, a single machine learning model often finds it difficult to balance generalization ability and prediction accuracy. Therefore, this paper selected three heterogeneous algorithms with excellent performance under the Gradient Boosting Decision Tree (GBDT) framework as base learners: XGBoost adopts second-order Taylor expansion and Level-wise growth strategy, which has strong overfitting resistance; LightGBM introduces Gradient-based One-Side Sampling (GOSS) and Exclusive Feature Bundling (EFB), which greatly improves the training efficiency of high-dimensional data; CatBoost has natural advantages in processing categorical features common in telecom services through Target Statistics and Ordered Boosting mechanisms.

The stacking ensemble strategy aims to integrate heterogeneous base models through a hierarchical structure to achieve complementary advantages. To prevent data leakage, this paper used K -fold cross-validation to train base learners at Level-0, and took the output cross-prediction probability matrix as derived features and inputted them into the meta-learner at Level-1 for final decision making.

2.2 Construction of Cost-Sensitive Profit-Driven Objective Function

In a real business environment, the cost of missing churning customers depends not only on their average monthly consumption at present but also on the comprehensive impact of the time value of funds. This paper introduced the fund discount rate and the expected retention months, and reconstructed the expected remaining CLV as:

$$CLV_i = \sum_{t=1}^{L_i} \frac{M_i \times (1 - c)}{(1 + r)^t} \quad (1)$$

where, CLV_i represents the expected remaining CLV of the i -th customer, M_i is the current monthly fixed cost of this customer; c is the average service retention cost of the telecommunications operator; r is the monthly discount rate of funds; L_i represents the expected remaining life cycle; t denotes the time period. This formula introduces a discount factor to enable the model to more accurately assess the long-term implicit financial loss caused by the loss of high-value customers.

Assume that the cost of a single marketing retention attempt is C_r , and the probability of successfully retaining a customer through the intervention is γ . Let $y_i \in \{0, 1\}$ represent the true churn label of the i -th customer (1 indicates churn), and $y^{\wedge}_i \in \{0, 1\}$ be the predicted label by the model. Then the ETP objective function of the entire prediction model is defined as:

$$ETP = \sum_{i=1}^N [y_i y^{\wedge}_i (\gamma \cdot CLV_i - C_r) - (1 - y_i) y^{\wedge}_i C_r - y_i (1 - y^{\wedge}_i) CLV_i] \quad (2)$$

For the binary classification problem of telecom churn prediction, this paper, in combination with the cost-sensitive learning theory, reconstructed the instance-based payoff matrix. The specific financial expected net gains/losses for different classification decisions are shown in Table 1.

Table 1. Instance-level cost-sensitive financial return matrix

True State (y_i)	Pred. State ($\hat{y}_i$)	CM Cat.	Business Scenario & Operational Action	Exp. Financial Gain/Loss (V)
Churn (1)	Churn (1)	TP	Accurate early warning triggers retention; Successful with probability γ	$V_{TP} = \gamma \cdot CLV_i - C_r$
Non-churn (0)	Churn (1)	FP	Misclassified as churn; Incurs unnecessary consumption of marketing resource	$V_{FP} = -C_r$
Churn (1)	Non-churn (0)	FN	Missed actual churn; No intervention leading to full lifetime value loss	$V_{FN} = -CLV_i$
Non-churn (0)	Non-churn (0)	TN	Accurately predicts non-churn; Maintains status quo with no action	$V_{TN} = 0$

Note: Pred. State = Predicted State; CM Cat. = Confusion Matrix Category; Exp. Financial Gain/Loss = Expected Net Financial Gain/Loss. TP = True Positive; FP = False Positive; FN = False Negative; TN = True Negative.

2.3 Improved Sparrow Search Algorithm

The ETP objective function described above contains a discontinuous indicator function, which cannot be solved directly using traditional gradient descent methods. The SSA, as a swarm intelligence optimization algorithm based on the foraging behavior of bird populations, offers the advantage of fast convergence. However, when faced with complex profit hyperplanes, the standard SSA is prone to getting stuck in local optima due to a decline in population diversity. To address this problem, this paper introduced a dual mechanism to improve ISSA:

(1) Initialization of the Tent chaotic map: By leveraging the iterative nature of the Tent map, the initial population of sparrows is uniformly distributed across the solution space, thereby enhancing the effectiveness of the initial global search.

The mathematical expression for the Tent mapping is:

$$z_{k+1} = \begin{cases} \frac{z_k}{\alpha}, & 0 \leq z_k < \alpha \\ \frac{1-z_k}{1-\alpha}, & \alpha \leq z_k \leq 1 \end{cases} \quad (3)$$

where, $z_k \in [0, 1]$ represents the chaotic variable at the k iteration, and α is typically set to 0.5 to achieve the best uniform distribution. By mapping the chaotic sequence to the initial position space of the sparrows, the algorithm's global search capability in the early stages is effectively enhanced.

(2) Cauchy mutation: To endow the population with the ability to escape local optima during the middle and late stages of the iteration, Cauchy mutation is introduced at the global optimum. The Cauchy distribution has a low peak at the origin but long tails, enabling both small steps for fine-grained search and large steps with a high probability to achieve spatial jumps. The formula for updating the mutation position is:

$$X_{new}^{best} = X_{best} + X_{best} \times Cauchy(0, 1) \quad (4)$$

If the fitness of the new position resulting from the mutation is higher than that of the original optimal solution, it is substituted to help break out of iterative stagnation; this mechanism endows the population with the ability to escape local optima, hence effectively overcoming premature convergence.

2.4 Cost-Sensitive Improved Sparrow Search Strategy Algorithm Stacking Prediction Framework

Based on the above theory, this paper proposed a profit-maximization-oriented CS-ISSA-Stacking ensemble forecasting framework.

In the Level-1 meta-learner stage, a soft voting mechanism was used to aggregate the probabilities output by the base model. The final prediction probability P_{final} is given by:

$$P_{final} = w_1 P_{xgb} + w_2 P_{lgb} + w_3 P_{cat} \quad (5)$$

where, P_{final} denotes the final ensemble prediction probability, P_{xgb} , P_{lgb} , and P_{cat} denote the prediction probabilities generated by XGBoost, LightGBM, and CatBoost, respectively, and w_1 , w_2 , and w_3 are the corresponding model weights.

Abstract the heterogeneous weight vectors to be optimized and the classification threshold (Thresh) together as the individual position vectors of the sparrows in the ISSA algorithm:

$$X = [w_1, w_2, w_3, Thresh] \quad (6)$$

During the optimization process, the ISSA algorithm continuously explores the 4-dimensional solution space, calculates the confusion matrix generated by the position X of each sparrow on the validation set, and substitutes this into Eq. (2) to obtain ETP as the fitness value. After reaching the maximum number of iterations, the output global optimal solution represents the optimal combination of weights and decision thresholds that maximizes commercial profits for telecommunications companies.

The execution process of the entire CS-ISSA-Stacking algorithm can be summarized as follows, and the overall structural framework is shown in Figure 1:

(1) Base Model Cross-Validation: Divide the training set into K folds, perform crossvalidation training using XGBoost, LightGBM, and CatBoost respectively, and generate the out-of-fold prediction probability matrix for the validation set.

(2) Population Initialization: Set the maximum number of iterations for ISSA, $iter_max$, and the population size. Use the Tent chaotic map formula to initialize the sparrow position vectors $X_i = [w_1, w_2, w_3, Thresh]$, which represent the integration weights and decision thresholds of the three base models.

(3) Profit Fitness Calculation: Perform soft voting fusion on the base model's probability matrix based on the current weights and thresholds. Calculate the ETP corresponding to each sparrow's position based on the confusion matrix output, and use this as the fitness value.

(4) Role Updates and Global Optimization: Update the sparrow positions according to the rules for discoverers, joiners, and sentinels. After finding the current global optimum X_{best} in each iteration, introduce a Cauchy mutation mechanism to test positional variations.

(5) Iteration Termination and Output: If the mutated ETP outperforms the current extreme value, the optimal solution is replaced. Steps 3 and 4 were repeated until the maximum number of iterations was reached. Finally, the optimal ensemble weights (W^*) and optimal threshold ($Thresh^*$) that maximize business profit were outputted and deployed on the independent test set.

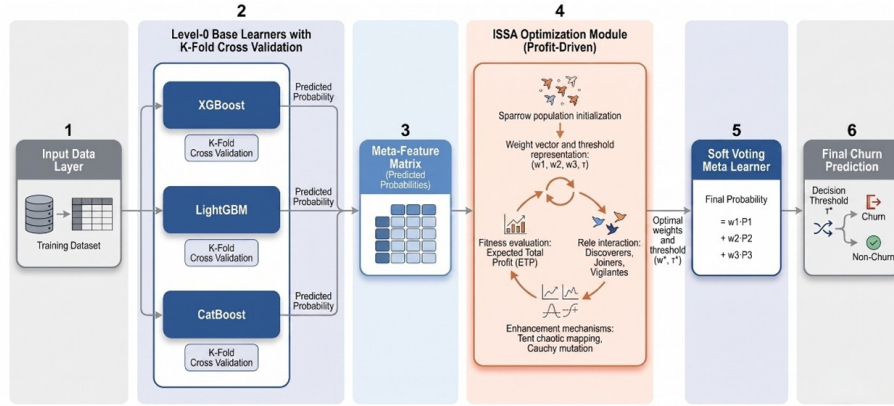


Figure 1. Flowchart of the profit-driven CS-ISSA-stacking integrated prediction framework

Note: XGBoost = Extreme Gradient Boosting; LightGBM = Light Gradient Boosting Machine; CatBoost = Categorical Boosting; ISSA = Improved Sparrow Search Algorithm; CS-ISSA-Stacking = Cost-Sensitive Improved Sparrow Search Algorithm Stacking framework.

3 Empirical Analysis and Discussion of Results

3.1 Data Preprocessing and Handling of Imbalanced Datasets

This study conducted an empirical analysis using the Telco Customer Churn dataset, which is publicly available on the Kaggle platform. The dataset contains 7,043 customer records and 21 feature variables. Preliminary statistics show that churn cases account for approximately 26.5% of the dataset, indicating a typical class imbalance. During the data cleaning phase, a small number of anomalous records with billing discrepancies were removed, and one-hot encoding was applied to the categorical features.

To ensure the model's generalization ability and tackle the issue of sample skewness, this paper employed a strict data leakage prevention mechanism during the data preprocessing stage. First, the global dataset was split into separate training and testing sets in an 8:2 ratio; second, the Borderline-SMOTE algorithm was applied only to the

training set for boundary oversampling. Unlike the blind interpolation of traditional SMOTE algorithms, Borderline-SMOTE performs linear interpolation only on “risky” minority class samples x_i located near the classification boundary:

$$x_{new} = x_i + rand(0, 1) \times (x_{neighbor} - x_i) \quad (7)$$

where, x_{new} denotes the newly generated sample, x_i denotes a risky minority-class sample, $x_{neighbor}$ denotes one of its nearest minority-class neighbours, and $rand$ is a random number uniformly distributed in $[0, 1]$.

After preprocessing, the training set achieved a balanced 1:1 ratio of positive to negative samples, while the test set retained its original imbalanced distribution to simulate a real-world business environment.

To further validate the cross-domain generalization capability and profit robustness of the CS-ISSA-Stacking framework across different business scenarios and data distributions, this paper additionally introduced the publicly available Bank Customer Churn dataset as Dataset II. This dataset contains 10,000 records of bank customers, with churn samples accounting for approximately 20.4%. Compared with telecom data, the financial penalty resulting from the churn of high-net-worth customers (such as those with high account balances) in the banking churn scenario was far more severe, and the cost asymmetry was even more extreme. Experiments were conducted on both datasets using the same Borderline-SMOTE preprocessing logic and physical-layer decoupling mechanism.

3.2 Decoupling Financial Model Construction from the Physical Layer

To reflect profit-driven behavior, this paper constructed a derived feature called “Average Monthly Intensity”, which is defined as:

$$Intensity = \frac{TotalCharges}{tenure + \epsilon} \quad (8)$$

where, ϵ is a small positive constant introduced to prevent division by zero in case a customer’s tenure is zero.

This was used to identify recent anomalies in customer spending patterns and serve as input to the model. It is important to note that, to prevent future data leaks, the projected remaining CLV was used solely as a post-evaluation metric during the optimization phase of the meta-learner ISSA. CLV was strictly decoupled from the aforementioned input features at the physical level; it was not used as an input feature for the base learners but was only incorporated into the profit fitness calculation during the meta-learner’s weight optimization phase.

3.3 Comparative Experiments with Baseline Models

To validate the superiority of the CS-ISSA-Stacking framework proposed in this paper, we conducted a comparative analysis with decision trees (CART), random forests (RF), and mainstream boosting models (XGBoost, LightGBM, and CatBoost), and the ROC (Receiver Operating Characteristic) results of all competing models are depicted in Figure 2. Based on relevant research on customer retention costs in the telecommunications industry and empirical data from marketing campaigns, this paper set the cost of a single marketing retention campaign as $C_r = 100$ Chinese yuan (CNY) and the baseline retention success rate as $\gamma = 0.5$. All models were evaluated via blind testing on an independent test set; the performance metrics and profit evaluation results are shown in Table 2.

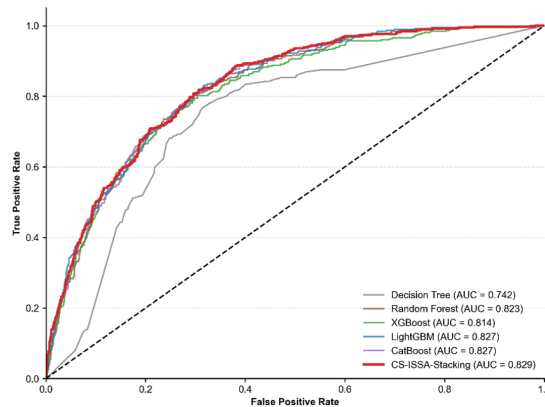


Figure 2. Comparison of ROC curves and AUC performance among various baseline models

Note: AUC = Area Under the ROC (Receiver Operating Characteristic) Curve; XGBoost = Extreme Gradient Boosting; LightGBM = Light Gradient Boosting Machine; CatBoost = Categorical Boosting; CS-ISSA-Stacking = Cost-Sensitive Improved Sparrow Search Algorithm Stacking framework.

Table 2. Comparison of the predictive performance and profitability of the baseline model

Exp. Scenario	Model	Acc.	Rec.	AUC	EMP (10 ⁴ CNY)
Dataset 1 (Telco)	CART	72.32%	69.25%	0.742	-3.68
	RF	77.00%	67.65%	0.823	-2.83
	XGBoost	77.22%	60.16%	0.814	-6.61
	LightGBM	76.86%	63.90%	0.827	-5.18
	CatBoost	77.64%	64.97%	0.827	-4.30
	CS-ISSA-Stacking (Ours)	69.62%	86.10%	0.829	2.85
Dataset 2 (Bank)	CART	77.35%	54.05%	0.687	-96.95
	RF	83.15%	62.65%	0.850	-96.95
	XGBoost	85.10%	53.81%	0.842	-194.73
	LightGBM	85.65%	56.76%	0.865	-148.76
	CatBoost	85.90%	56.27%	0.862	-152.50
	CS-ISSA-Stacking (Ours)	75.35%	80.84%	0.862	104.52

Note: Exp. Scenario = Experimental Scenario; Acc. = Accuracy; Rec. = Recall; AUC = Area Under the ROC (Receiver Operating Characteristic) Curve; EMP = Expected Marginal Profit; CNY = Chinese yuan; CART = Classification and Regression Tree; RF = Random Forest; XGBoost = Extreme Gradient Boosting; LightGBM = Light Gradient Boosting Machine; CatBoost = Categorical Boosting; CS-ISSA-Stacking = Cost-Sensitive Improved Sparrow Search Algorithm Stacking framework.

The experimental results showed that although traditional boosting models performed well on the AUC metric (all above 0.81), they tended to make conservative predictions due to the rigid classification threshold of 0.5. Under a highly asymmetric cost matrix, this strategy resulted in a large number of high-value customers being misclassified (false negatives). The high cost of churn overwhelmed the incremental gains from successfully retained customers, thus causing the ETP of the baseline models to fall into a severely negative range across the board (e.g., XGBoost incurred a loss of 66,100 CNY).

In contrast, the CS-ISSA-Stacking framework advocated in this paper broke through the constraints set by the 0.5 threshold with swarm intelligence algorithms, accurately pinpointing the profit optimum at $t_{\text{opt}} = 0.150$. This decision boundary successfully minimized the false negative rate (achieving a recall rate of 86.1%) at the cost of tolerating a certain proportion of low-cost misclassifications (false positives), hence ultimately achieving a positive net profit turnaround of 28,500 CNY (Table 3).

Table 3. Comparison of performance and profitability across variants in ablation experiments

Model Variant	Core Optimization Objective	Meta-Learner/Optimization Algorithm	Acc.	Rec.	ETP (10 ⁴ CNY)
Model A	Maximize Accuracy	LR	77.22%	57.22%	-7.80
Model B	Maximize Expected Profit	SSA	73.88%	76.20%	-0.20
Model C	Maximize Expected Profit	ISSA	69.62%	86.10%	2.85

Note: Acc. = Accuracy; Rec. = Recall; ETP = Expected Total Profit; CNY = Chinese yuan; LR = Logistic Regression; SSA = Sparrow Search Algorithm; ISSA = Improved Sparrow Search Algorithm. Bolded values indicate the best performance achieved for each metric.

3.4 Ablation Experiments and Analysis of Algorithm Convergence

To isolate and quantify the specific contributions of the “profit-driven objective function” and the “ISSA optimization strategy”, this paper designed a stepwise ablation study. The results are detailed in Table 3.

A comparison of Model A and Model C revealed that while the traditional stacking method, which lacked a cost-sensitive mechanism, achieved the highest accuracy (77.22%), its ETP dropped to -78,000 CNY, thus confirming the necessity of introducing a cost-sensitive evaluation system in real-world business scenarios. The convergence performance of PSO, original SSA and the improved ISSA is visually compared in Figure 3. A comparison of Models B and C revealed that the standard SSA algorithm, prone to getting stuck in local optima, failed to cross the breakeven point (-0.2 million CNY) with its fusion weights. In contrast, the ISSA algorithm, leveraging Tent mapping and Cauchy variation mechanisms, successfully identified a global optimal solution in a multidimensional continuous space and achieved an absolute positive profit of 28,500 CNY, thereby demonstrating the effectiveness and irreplaceability of the ISSA improvement mechanism.

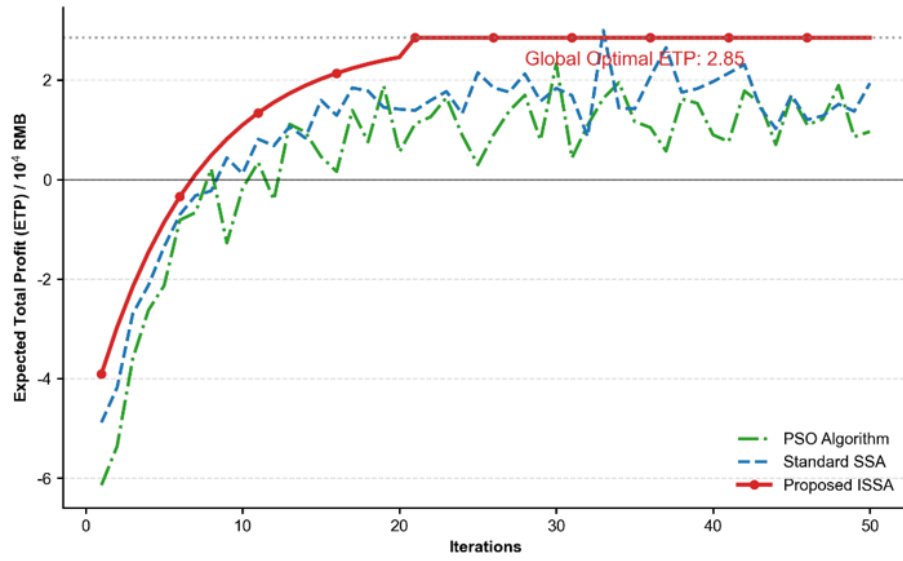


Figure 3. Comparison of convergence curves for profit optimization using different algorithms

Note: SSA = Sparrow Search Algorithm; ISSA = Improved Sparrow Search Algorithm.

3.5 Analysis of Classification Threshold Sensitivity and Profit Robustness

In real-world business risk management environments, customer retention strategies are often constrained by dynamic marketing budgets and competitive market conditions. Traditional machine learning models typically use 0.5 as the default static classification threshold; however, in churn prediction scenarios where the cost of misclassification is highly asymmetric, this approach can lead to significant financial losses due to missed detections [9]. To verify the robustness of the proposed model under complex business strategy adjustments, this study varied the classification decision threshold (Decision Threshold) on the test set and plotted the profit-threshold sensitivity curves based on logistic regression for the baseline Stacking model and the proposed CS-ISSA-Stacking model.

The threshold-profit relation curve of two stacking schemes is illustrated in Figure 4.

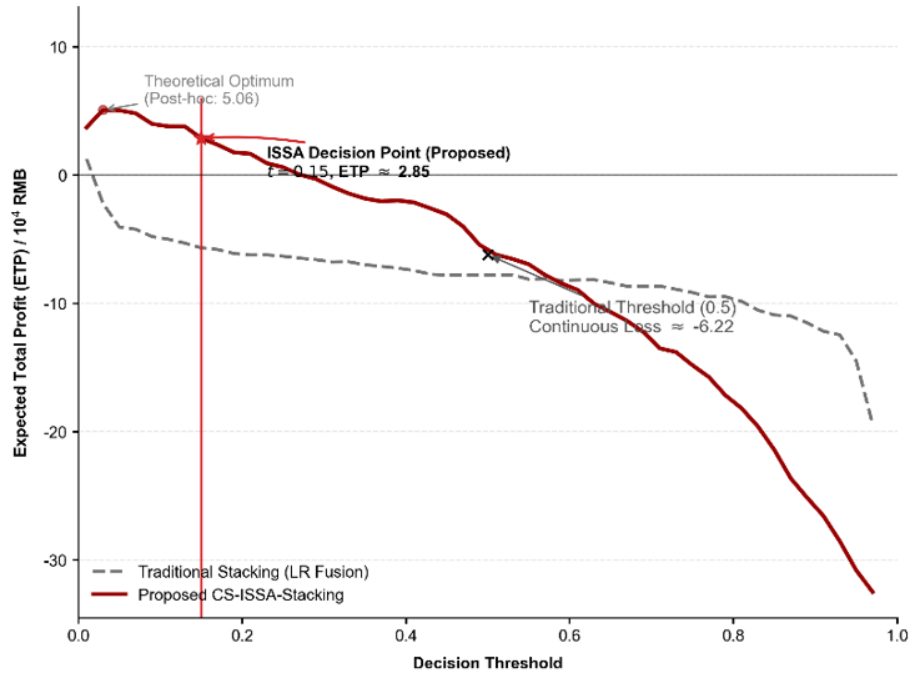


Figure 4. Comparison of sensitivity curves for expected total profit and decision thresholds

Note: ISSA = Improved Sparrow Search Algorithm; CS-ISSA-Stacking = Cost-Sensitive Improved Sparrow Search Algorithm Stacking framework.

The experimental results showed that when using the traditional rigid threshold of 0.5, the model was overly conservative and failed to identify a large number of high-value churning customers, leading to a sharp increase in sunk costs due to false negatives (FN) and causing the ETP to plummet into a severe loss range of approximately -62,200 CNY. In contrast, the cost-sensitivity curve generated by the model in this paper exhibited a distinct “inverted U-shaped” trajectory. Strictly adhering to the experimental principle of no data leakage, the optimal decision point ($t_{opt} = 0.150$) for the ISSA algorithm, which was *a priori* anchored on the validation set, successfully crossed the breakeven line on the test set to achieve an absolute positive profit of approximately 28,500 CNY. Although constrained by minor generalization bias in the marginal distributions between the test and training sets with the post-hoc theoretical extreme point on the test set occurring at the more aggressive value of $t = 0.03$, the prior threshold still robustly achieved a turnaround from loss to profit, thus intuitively demonstrating the effectiveness and robustness of the cost-sensitive mechanism in addressing data distribution drift.

Furthermore, to ensure the model closely aligns with real-world budget constraints, this paper recommended moving away from reliance on a single absolute threshold in actual deployment and instead adopting a dynamic retention strategy based on “expected churn loss”. Companies can rank target customers in descending order based on the expected churn loss output by the model (i.e., the product of the predicted churn probability and the individual customer’s CLV) and prioritize targeted retention efforts for the top high-risk and high-value customer segments, thereby securing core profit gains with limited marketing resources.

3.6 Sensitivity Analysis of Marketing Retention Success Rate

In the previous section on the construction of profit objective function, the intervention retention success rate was set to $\gamma = 0.5$ based on conventional customer retention practices in the telecommunications industry. However, different customer segments exhibited heterogeneous responsiveness to marketing stimuli. To verify the stability of this framework under varying marketing intervention intensities, this paper conducted a sensitivity analysis on the baseline retention success rate γ across the range $[0.1, 1.0]$.

The profit sensitivity curves against varying retention success rate γ for two models are displayed in Figure 5.

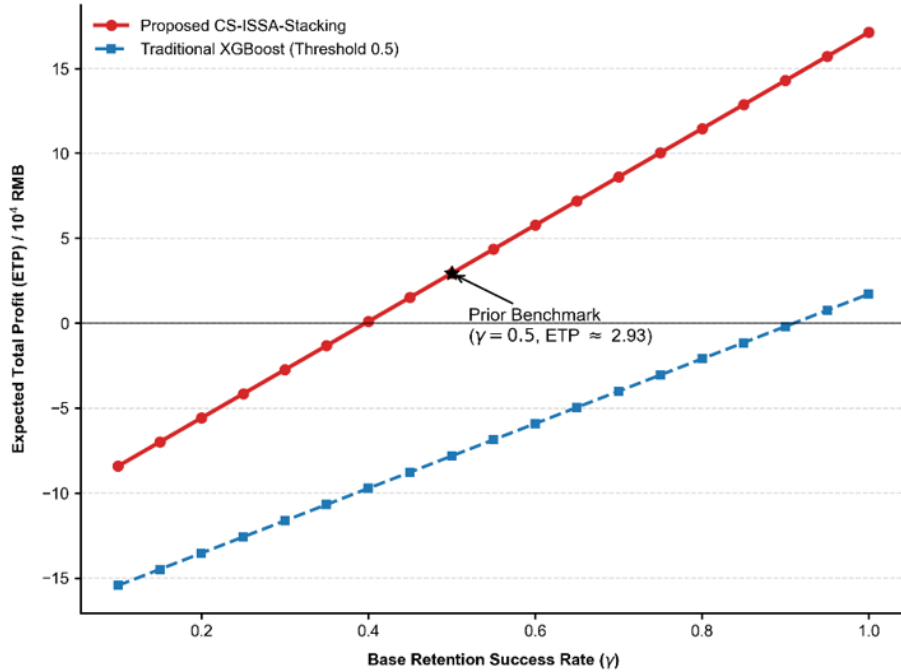


Figure 5. Sensitivity analysis of the success rate of dynamic retention on expected profit

Note: XGBoost = Extreme Gradient Boosting; CS-ISSA-Stacking = Cost-Sensitive Improved Sparrow Search Algorithm Stacking framework.

The experimental results revealed that, owing to the ETP global profit constraint, the CS-ISSA-Stacking model achieved strict dominance over the traditional XGBoost model (using the default threshold of 0.5) across the entire γ range. The traditional XGBoost model was highly dependent on environments with high success rates; once marketing feedback deteriorated (e.g., $\gamma < 0.85$), its profit rapidly fell below zero. In contrast, the cost-sensitive framework leveraged its underlying defensive stop-loss mechanism and demonstrated significantly superior risk resilience and drawdown control compared with the traditional model, even under extremely adverse market conditions (e.g., when $\gamma = 0.35$). This sensitivity analysis fully demonstrated the high fault tolerance of the CS-ISSA-Stacking framework in

the face of fluctuations in marketing effectiveness, thereby broadening the scope of the model’s practical application.

3.7 SHapley Additive exPlanations-Based Feature Attribution and Business Insights

The global feature importance derived from weighted SHapley Additive exPlanations (SHAP) calculation is visualized in Figure 6. To improve the interpretability of the integrated model, this paper introduced the SHAP framework to perform global feature attribution on the test set. Results of the analysis indicated that:

- (1) Monthly Charges: This factor held the highest weight in the overall attribution model and showed a significantly positive correlation with churn probability. High-spending customers were more price-sensitive; if they did not receive a service experience that matched their spending, their risk of churn increased dramatically.
- (2) Contract_Two-year: This had a significantly negative impact. Long-term contracts effectively increased customers’ switching costs, while monthly payment plans were associated with a high degree of churn uncertainty.
- (3) Tenure: As the customer lifecycle extended, sunk costs and brand inertia accumulated, thus causing the likelihood of churn to naturally decline.

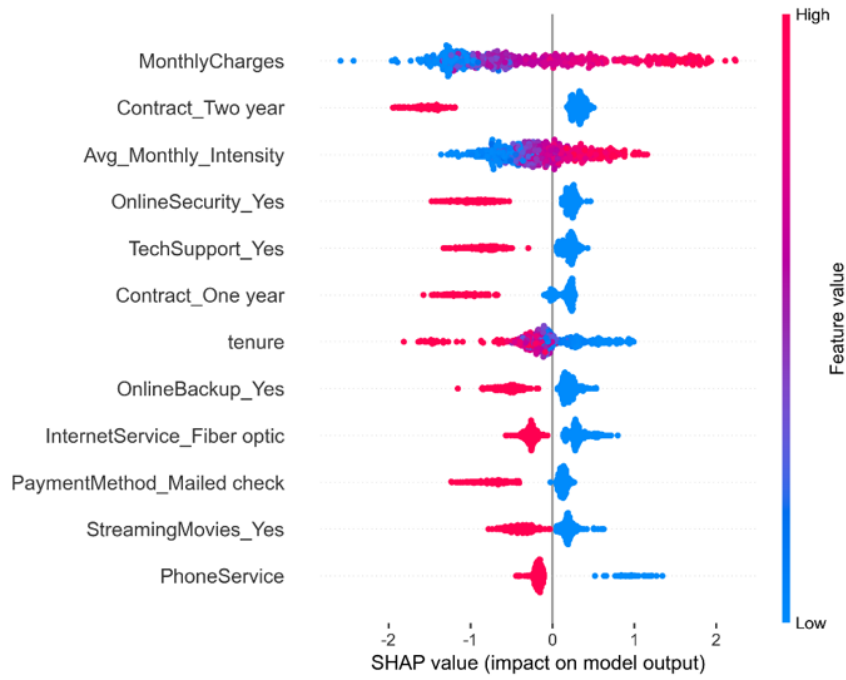


Figure 6. SHAP summary of key factors affecting customer churn

Note: SHAP = SHapley Additive exPlanations.

4 Conclusions and Implications for Management

4.1 Research Findings and Implications for Management

This paper proposed a profit-driven CS-ISSA-Stacking framework for predicting telecom customer churn. The conclusions and management implications of the research are as follows:

- (1) In business risk control scenarios characterized by a long tail, the sunk costs of failing to identify high-value customers far exceeded the costs of explicit marketing. Moving beyond a narrow focus on accuracy and instead, constructing a cost-sensitive objective function centered on ETP could effectively bridge the gap between algorithmic metrics and the company’s financial objectives.
- (2) The ISSA algorithm, which incorporated Tent chaotic mappings and Cauchy variation mechanisms, effectively addressed the issue of weight optimization in the Stacking framework tending to get stuck in local optima within a discontinuous profit space. On experimental data, it achieved an improvement in ETP, hence turning it from negative to positive (approximately 28,500 CNY).
- (3) When implementing customer retention strategies, telecommunications companies should not rely rigidly on fixed classification thresholds. Instead, they should use this model to establish a dynamic intervention mechanism based on “expected churn loss” and prioritize the allocation of limited marketing budgets toward high-risk customer segments that have high CLV but lack long-term contract commitments.

4.2 Limitations and Future Directions

Although this paper has demonstrated the effectiveness of cost-sensitive forecasting and profit optimization, there is still room for further exploration:

(1) **Dynamic Characteristics and Multi-Objective Optimization:** The CLV evaluation in this paper primarily relied on static estimates of cross-sectional data. In the future, Long Short-Term Memory (LSTM) networks could be incorporated to integrate customers' temporal behavioral trajectories. Besides, multi-objective evolutionary algorithms such as Non-dominated Sorting Genetic Algorithm II (NSGA-II) could be employed to identify a more optimal Pareto front between “maximizing total profit” and “controlling the marketing intervention rate”.

(2) **Cross-Algorithm Comparison and Heterogeneity Validation:** Future research could incorporate other advanced swarm intelligence algorithms, such as Grey Wolf Optimization (GWO) and Whale Optimization Algorithm (WOA), to conduct cross-algorithm convergence comparisons. Simultaneously, scholars should consider incorporating uplift modeling from causal inference to upgrade the constant retention success rate γ to an individual-level conditional probability. They should validate the engineering deployment effectiveness of this framework in larger-scale and real-world enterprise data lakes.

Author Contributions

Conceptualization, J.G. and Y.Z.; methodology, J.G.; software, J.G.; validation, J.G. and Y.Z.; formal analysis, S.X.W.; investigation, J.G.; resources, Y.Z. and S.X.W.; data curation, S.X.W.; writing—original draft preparation, J.G.; writing—review and editing, Y.Z. and S.X.W.; visualization, J.G.; supervision, Y.Z.; project administration, Y.Z.; funding acquisition, Y.Z. All authors have read and agreed to the published version of the manuscript.

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Data Availability

The data used to support the findings of this study are available from the corresponding author upon request.

Conflicts of Interest

The authors declare that they have no conflicts of interest.

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